

What Architecture Should I Choose for my Continuous-Time Delta-Sigma Modulator?

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Abstract—A novice continuous-time delta-sigma designer is faced with an admittedly complex maze of possible design choices. The right architecture often determines how efficiently the modulator can be implemented. This paper critically examines various popular delta-sigma architectures. It concludes that a single-bit modulator with FIR feedback is a prime candidate that enables a power-efficient implementation for a variety of specifications. To support this thesis, measurement results of an audio delta-sigma modulator, designed in a 65 nm CMOS process are given. The modulator, which incorporates FIR feedback and chopping to reduce $1/f$ noise, achieves 98.6 dB peak SNDR in a 24 kHz bandwidth and consumes only 260 μ W from a 1.2 V supply.

I. INTRODUCTION

Continuous-time $\Delta\Sigma$ modulation is now a well-established analog-to-digital conversion technique, that has not only been used in high-precision low-bandwidth applications, but also to digitize signals with bandwidths in the 100's of MHz range. CT $\Delta\Sigma$ s are easy to drive, and have relaxed requirements on active elements (when compared to their discrete-time counterparts). Further, they have the property of *inherent antialiasing*, which makes them robust to substrate noise when embedded in large digital chips. The gamut of applications using CT $\Delta\Sigma$ s has been increasing, and is expected to grow further as advances in CMOS technology enable higher speeds of operation.

The block diagram of a CT $\Delta\Sigma$ is shown in Fig. 1. The input u and the feedback DAC waveform $v(t)$ are processed by a loop filter, whose output is sampled, quantized and fed back to the DAC. There are many “free variables” that can result in the same in-band SNR. For example, the number of quantizer levels can be reduced at the expense of a higher sampling rate; or the order of the loop filter can be increased, while operating at a lower sampling rate; or the order and sampling rate can be reduced while increasing the number of levels. Add to this the many possible styles to realize integrators, ADCs and DACs, a designer has the unenviable job of having to pick from a dizzying array of possibilities.

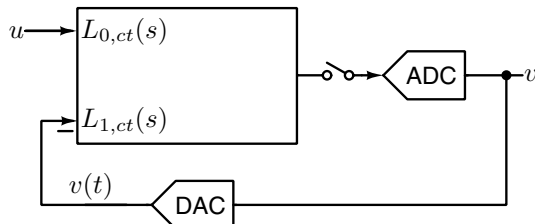


Fig. 1. Block diagram of a CT $\Delta\Sigma$.

The aim of this paper is to help the designer navigate the

“architectural maze”. In view of space constraints, we consider “low-bandwidth” designs; where the speed of the technology allows very large oversampling ratios (OSR), so that the designer is free to choose an arbitrarily high sampling frequency. The situation above is representative of many practical situations – for example, an audio CT $\Delta\Sigma$ can be operated at (almost) any OSR even in a legacy CMOS technology. Similar arguments hold for designs targeting sensor interfaces and Internet of Things (IoT). The question we wish to answer, therefore, can be recast as “given a technology which is fast enough, what is the most power efficient way of implementing a CT $\Delta\Sigma$?” This paper, which compares various choices for the design of a single-loop CT $\Delta\Sigma$, is organized as follows. In Section II we describe the pros and cons of single-bit and multibit CT $\Delta\Sigma$ s and reason that a single-bit CT $\Delta\Sigma$ with an FIR feedback DAC is an attractive candidate for a variety of specifications. It turns out that the performance of this architecture holds up well across process technologies. To support this, Section III gives experimental results of an audio CT $\Delta\Sigma$ designed in a 65 nm CMOS process. This design uses the same architectural and circuit design techniques as the design in [1], which used a 180 nm CMOS process. The CT $\Delta\Sigma$ reported in this paper achieves a peak SNDR of 98.6 dB in a 24 kHz bandwidth, while consuming only 260 μ W from a 1.2 V supply. The Schreier Figure-of-Merit (FoM) is the highest reported for an audio CT $\Delta\Sigma$. Section IV concludes the paper.

II. ARCHITECTURAL COMPARISON

The number of levels in the quantizer plays a crucial role in determining power dissipation, irrespective of other factors. Our first task at hand, therefore, is to discuss considerations for the design of the quantizer (which is the cascade of an ADC and a DAC).

A. ADC Design

The ADC in a one-bit modulator is simple to implement, as it consists of a single comparator. Since the ADC compares the loop-filter output with zero, a reference ladder is not necessary. Clock generation and distribution circuitry is simple, and consumes relatively less power. More importantly, comparator offset is of no consequence. This means that small devices can be used in the comparator, reducing its power dissipation, and that of the clock generation circuit [2].

In a multibit design, the ADC is commonly implemented using a flash architecture so as to minimize delay in the loop. As a result, the hardware complexity of the ADC increases

exponentially with the number of bits resolved. Though the sampling rate needed to achieve a desired SQNR in a multibit modulator is lower, the increased number of comparators more than compensates for the reduction in operating frequency. For example, using a 4-bit quantizer instead of a 1-bit one in a third order CT $\Delta\Sigma$ enables the sampling rate to be reduced by a factor of about 2. This means that if the *same* comparator was used in both the single-bit and multi-bit designs, the ADC in the latter would consume about $8\times$ more power though operating at half the rate. Unfortunately, it also turns out that the comparators used in the multibit case need to be “better”. The reason is that comparator offsets increase the variance of quantization noise injected by the ADC, and reduce the maximum stable amplitude of the modulator [3].

A multibit loop also complicates the design of the loop filter due to the following. The input range of the ADC in a multi-bit loop is often chosen to be as large as possible to relax offset requirements on the comparators of the flash ADC. This means that the swing at the output of the loop-filter has to be large, which makes the design of the integrator driving the flash ADC difficult. Combined with the fact that a multi-bit ADC presents a larger load to the loop-filter, the design of the loop filter can become particularly challenging. The single-bit CT $\Delta\Sigma$ scores over its multi-bit cousin on this count. After all, the 1-bit ADC only determines the sign of the loop-filter’s output, which means that the latter can be conveniently scaled without affecting the modulator’s output. All things considered, therefore, it appears as if the design of the ADC in a multibit modulator is *much* more involved, consumes a larger area and dissipates more power.

B. DAC Design

The performance of a negative feedback loop is a strong function of its feedback block, and a CT $\Delta\Sigma$ is no exception to this rule. The fidelity of the DAC, therefore, has a tremendous bearing on its in-band performance. Distortion in the DAC is of two types – static and dynamic. The former is caused by element mismatch in the DAC. The latter is dependent on implementation details. Assuming an NRZ DAC, dynamic distortion is due to inter-symbol-interference (ISI) caused by unequal rise and fall times of the DAC pulse.

A commonly heard refrain is that “the two level DAC needed in a single-bit modulator is inherently linear”. This is largely true, though it neglects dynamic distortion introduced by the DAC, which limits linearity at high speeds/resolution.

Element mismatch in the feedback DAC of a multi-bit CT $\Delta\Sigma$ can dramatically degrade the in-band SNDR. This has spawned a variety of methods that address the problem – ranging from analog calibration to dynamic element matching (DEM). Most of these techniques address only static mismatch. DEM, in particular, is a hugely popular technique that shapes mismatch noise out of the signal band. This, however, introduces additional delay in the feedback loop and increases power dissipation. Further, DEM algorithms designed without dynamic mismatch in mind, in combination with DAC designs that do not carefully address this issue, typically *worsen*

CT $\Delta\Sigma$ performance. While recent DEM algorithms address dynamic distortion, it appears as if the design complexity is increased. In summary, dynamic distortion is a problem with single-bit and multi-bit modulators, and needs to be addressed anyway. From the implementation viewpoint, therefore, a single-bit DAC seems a lot more attractive.

C. Clock Jitter

From the discussion in the preceding subsections, we saw that a single-bit quantizer results in a power and area efficient implementation. However, the price to be paid for this simplicity is an increased sensitivity to clock jitter (when compared to a CT $\Delta\Sigma$ using a multibit quantizer). The root cause of this is the large steps in the 1-bit feedback DAC waveform. Fortunately, this can be addressed using FIR feedback [4], [5], as shown in Fig. 2. The 2-level output

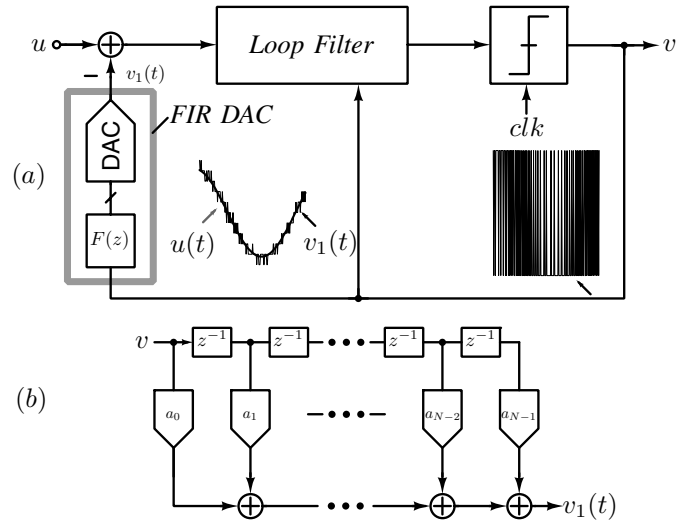


Fig. 2. Using an FIR feedback DAC to mitigate noise due to clock jitter in a single-bit modulator.

sequence v is filtered by an N -tap low pass FIR filter with transfer function $F(z)$, before exciting the main feedback DAC. The DAC has an NRZ pulse shape. For simplicity, the tap weights of $F(z)$ are assumed to be identical. Since $v = \pm 1$, the magnitude of transitions in v is 2. With $F(z)$ in place, this is reduced to $2/N$. Thus, the magnitudes of the steps in the feedback DAC waveform (denoted by $v_1(t)$ in Fig. 2(a)) are N times smaller than what they would have otherwise been. Since noise due to clock jitter is proportional to the height of the transitions in the DAC output, it follows that the in-band mean square noise due to jitter is reduced by $20\log(N)$ dB.

The FIR DAC has other important benefits. Since $F(z)$ is a low pass filter, the input component of v is not affected, but the power of the shaped noise is reduced. The DAC’s output $v_1(t)$, therefore, has a reduced high frequency content that closely follows the input u . The error processed by the loop filter, which is $u(t) - v_1(t)$, is therefore much smaller. This relaxes the linearity requirements of the loop filter, just like in a CT $\Delta\Sigma$ with a multi-bit DAC. Intuitively, since $v_1(t)$ “looks” like the output of a multibit DAC, one should

expect similar benefits with respect to clock jitter and loop filter linearity.

Implementing the FIR DAC as drawn in Fig. 2(a) is problematic when the DAC levels are not equally spaced (due to mismatched components). Recognizing that $v[n]$ is a two level sequence, a linear DAC-filter combination can be realized using the semi-digital approach [6], as shown in Fig. 2(b). Here, the delays are implemented digitally, while the individual DAC outputs (which are assumed here to be currents) are weighted and summed in the analog domain. It is easy to see that DAC mismatch modifies the transfer function of the filter, but does not cause nonlinearity. Using a single-bit modulator with FIR feedback (originally intended to address clock jitter) also helps with respect to the linearity of the loop filter, as seen from the input and feedback waveforms of Fig. 2.

The FIR DAC introduces delay into the $\Delta\Sigma$ loop and can result in instability. Fortunately, the loop is easily stabilized using the method of moments [7].

To summarize the discussion so far, we have seen that using a plain 1-bit quantizer, while being a very efficient use of hardware, is severely impacted by jitter and is problematic for the linearity of the CT $\Delta\Sigma$. Fortunately, using FIR feedback with a one-bit ADC not only addresses the jitter problem, but also greatly relaxes the linearity requirements of the loop filter, just like with a multibit quantizer. FIR feedback, therefore, combines the beneficial aspects of single bit and multibit designs, without the disadvantages of either approaches. The benefits of FIR feedback hold up well even in advanced technologies, as described in the next section.

III. PROCESS ROBUSTNESS

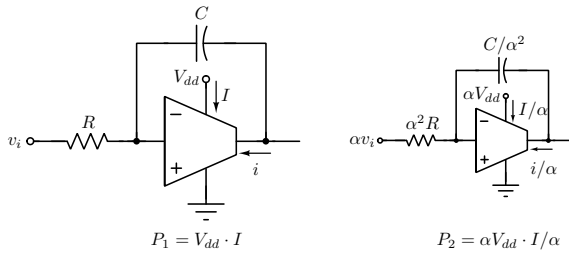


Fig. 3. Power dissipation needed to achieve the same SNR over a given bandwidth does not change with technology scaling.

In principle, power dissipation of a CT $\Delta\Sigma$ that achieves a given SNR over a fixed bandwidth should not depend on the supply voltage, as explained using Fig. 3. On the left of the figure is shown an OTA-RC integrator operating with a supply voltage V_{dd} . The peak current sourced/sunk and the current drawn by the OTA are denoted by i and I respectively. The power dissipated is $P_1 = V_{dd}I$. In a scaled technology, where the supply voltage is αV_{dd} , the peak input signal swing possible is also scaled by α . To achieve the same SNR therefore, the input resistors must be scaled by α^2 . The peak current sourced/sunk by the OTA will now increase to i/α . As a result, the quiescent current drawn by the OTA should now be I/α . The power dissipation is thus given by $P_2 = \alpha V_{dd} \cdot (I/\alpha) = P_1$.

In practice, however, advanced processes operate with lower supply voltages, and have transistors with poor analog performance (g_m/g_{ds}) and large flicker-noise corners. It is thus commonly believed that low-bandwidth high-resolution CT $\Delta\Sigma$ architectures that achieve excellent power-efficiency in older process nodes will not remain as efficient when migrated to sub-100nm CMOS technologies. In light of this, it is interesting to examine if the benefits of FIR feedback hold up with technology scaling. It is also desirable that no new “tricks” need to be invented to obtain the performance that was achieved in the older process technology.

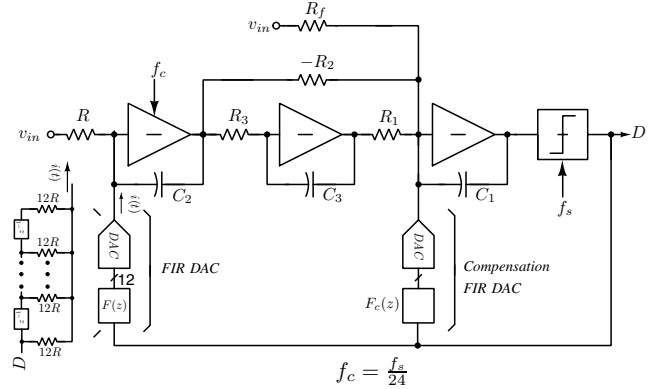


Fig. 4. Architecture of the CT $\Delta\Sigma$ designed in a 65 nm CMOS process. It is the same as that in [1].

To demonstrate the robustness of the FIR-DAC architecture with process and supply scaling, the CT $\Delta\Sigma$ of [1] (implemented in a 180nm CMOS process with a 1.8 V supply) was ported to a 65nm CMOS technology with $V_{dd} = 1.2$ V. The design is simply scaled using the principle of Fig. 3. The analog power should, therefore, be expected to be substantially the same as in the 180 nm design. The digital power, on the other hand, should reduce due to lower voltage operation and the advanced technology node. The design details remain largely the same as in [1] and are omitted here. As in that design, the input integrator is chopped at $f_s/24 = 256$ kHz. The use of chopping is of key importance in the 65 nm design due to larger $1/f$ noise. We focus on comparing the relative performance of the two designs.

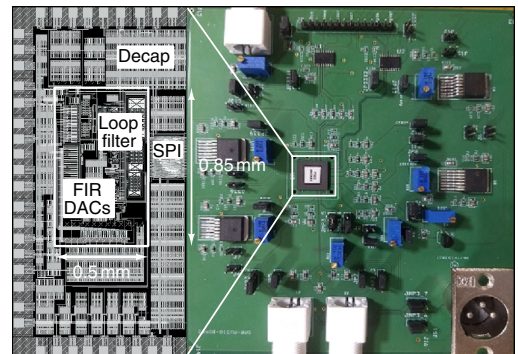


Fig. 5. Layout snapshot and photograph of the test board.

Fig. 5 shows a layout snapshot of the 65 nm test chip and a photograph of the board used for characterization. When

compared to [1], the area is smaller by a factor of about 2.5. This is due to the following. A large part of the area is due to the resistive FIR DAC. Smaller resistors needed in the 65 nm design will reduce area.

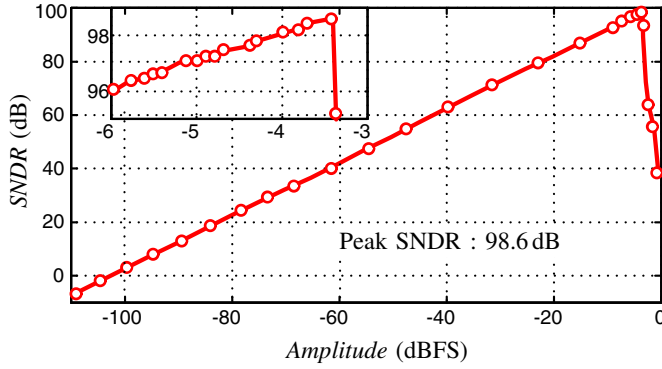


Fig. 6. Measured SNDR as a function of signal amplitude. The input frequency is 6 kHz. The dynamic range is 103 dB.

Fig. 6 shows the measured SNDR as a function of the input amplitude. The peak SNDR is 98.6 dB, about the same as that achieved in [1]. The dynamic range is about 103 dB. Fig. 7 shows the power spectral density (PSD) for a -3.4 dBFS input. We see that HD_2 and HD_3 are about 120 dB. The lower distortion can be attributed to the reduced effect of inter-symbol interference (ISI) in the feedback DAC due to the substantially smaller rise/fall times in the 65 nm process.

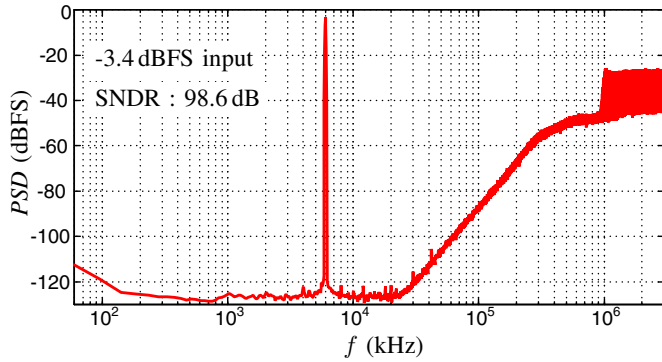


Fig. 7. Measured PSD for a -3.4 dBFS tone at 6 kHz. The in-band SNDR is 98.6 dB.

Fig. 8 shows the measured PSD with/without chopping, for a -6 dBFS input at 6 kHz. Without chopping, the $1/f$ noise corner is greater than the signal bandwidth! Thanks to chopping, the flicker noise corner is about 10 Hz, like in [1]. To achieve the same flicker noise corner without chopping would necessitate an input stage which is about 10000 times larger.

The 65 nm design dissipates 260 μ W from a 1.2 V supply. This is marginally smaller than the 180 nm design in [1], and is mainly due to the reduced current-draw of the digital circuitry. Table I compares the performance of the 65 nm design reported in this work to some state of the art modulators. Barring [1], all the designs are implemented in processes with a minimum feature size of 65 nm or lower. We see that our design achieves a substantially higher Schreier figure-of-merit (FoM) than [8]–[10], largely attributable to the FIR feedback architecture.

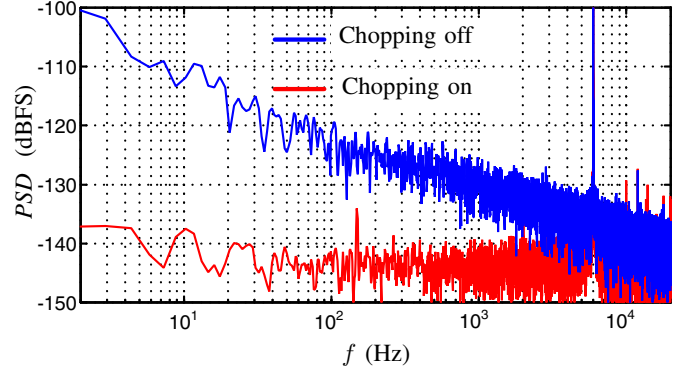


Fig. 8. Measured PSD for a -6 dBFS input with/without chopping.

When compared to [1], the Schreier FoM is 0.5 dB higher, demonstrating that the benefits to be had with FIR feedback hold up well even in advanced low-voltage CMOS processes.

TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART $\Delta\Sigma$ CONVERTERS.

Reference	[8]	[9]	[10]	[1]	This work
BW (kHz)	24	24	25	24	24
Node (nm)	65	28	65	180	65
Supply (V)	1.1	3.3/1.0	1.0	1.8	1.2
Power (μ W)	121	1130	800	280	260
Peak SNDR (dB)	85	98.5	95.2	98.5	98.6
SFDR (dB)	90	102.6	100	107.6	120
Area (mm^2)	0.6	0.02	0.25	1	0.42
Chop freq. (kHz)	-	-	47	256	256
FoM _{Schreier}	168	172	170	177.8	178.3

$$\text{FoM}_{\text{Schreier}} = \text{SNDR}_{\text{max}} + 10 \log_{10} \left(\frac{\text{BW}}{\text{Power}} \right)$$

IV. CONCLUSIONS

We compared the relative power efficiencies of CT $\Delta\Sigma$ Ms employing single-bit, multibit and single-bit+FIR feedback. A low-resolution quantizer, used in conjunction with FIR feedback, not only simplifies the design of the quantizer, but also reduces the sensitivity of the modulator to clock jitter. FIR feedback, therefore, combines the benefits of single-bit and multi-bit operation. Another useful benefit of FIR feedback, particularly in CT $\Delta\Sigma$ Ms targeting low-bandwidth, is that it allows the OTA of the input integrator to be chopped “for free”. The power efficiency of FIR feedback holds up well with technology and supply-voltage scaling. We demonstrated this with experimental results from an audio $\Delta\Sigma$ modulator that achieves 98.6 dB SNDR and 120 dB SFDR in a 24 kHz bandwidth. Operating from a 1.2 V supply in a 65 nm CMOS process, the CT $\Delta\Sigma$ M consumes only 260 μ W, and achieves a Schreier (SNDR) FoM of 178.3 dB, which is 0.5 dB higher than that of the design of [1] (which used a 180 nm process and a 1.8 V supply).

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